Title: Participatory Design and Design for Values

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Abstract:

Participatory Design (PD) is a design methodology in which the future users of a design participate as co-designers in the design process. It is a value-centred design approach because of its commitment to the democratic and collective shaping of a better future. This chapter builds forth on the Scandinavian Participatory Design tradition. We discuss why the design process is as important as the final result, the product or service. The creative application of participatory design methods facilitates a design process in which values emerge and become inscribed in a prototype. We present PD's guiding principles: equalising power relations, democratic practices, situation-based action, mutual learning, tools and techniques, and alternative visions about technology. In addition we discuss some value practices and design methods informed by our PD projects in healthcare and the public sector. We maintain that Participatory Design increases the chance that the final result of a design process represents the values of the future users.

Participatory Design and Design for Values

1. Introduction

Participatory Design (PD)[1] is a collection of design practices for involving the future users of the design as co-designers in the design process. PD's methodology is based on the genuine decision-making power of the co-designers and the incorporation of their values in the design process and its outcome, which is often a high-fidelity prototype for a product or service, or a new way to organise a work practice or to design a space. The core theme of Participatory Design, as formulated by Pelle Ehn (1988), is addressing the dialectics of tradition and transcendence: the tension between *what is* and *what could be*. PD's methods enable participants to anticipate future use and alternative futures.

In the most recent comprehensive volume of writings about Participatory Design (Simonsen and Robertson, 2012), PD is defined as follows:

> A process of investigating, understanding, reflecting upon, establishing, developing, and supporting mutual learning between multiple participants in collective 'reflection-in-action'. The participants typically undertake the two principle roles of users and designers where the designers strive to learn the realities of the users' situation while the users strive to articulate their desired aims and learn appropriate technological means to obtain them (p. 2).

The Participatory Design tradition was established in Scandinavia in the early 1970s. It was influenced by, and developed concurrently, with a range of projects with a focus on the democratisation of the working life. These projects were conducted by the trade unions or jointly by the trade unions and working life researchers. The Norwegian Iron and Metal Workers union (NJW) initiated one of the first PD projects in cooperation with researchers from the Norwegian Computing Centre from 1970 to 1973. The objective was to involve the workers in the design of a computer-based planning and control system for their workplace. A plan was designed, based on a participative approach and the inclusion of workers' knowledge, with several activities for the unions, including working groups to discuss and to find solutions through action programs, assessments of existing information systems, and propositions of changes. The researchers participated with lectures as well as with support in the development of the project results (Nygaard and Bergo, 1975). In addition, educational material, in the form of a textbook, was developed to support these activities (Nygaard, 1974).

Similar projects followed in other sectors, such as the Swedish project *Democratic Control and Planning in Working Life* (DEMOS)[2] (Ehn and Sandberg, 1979) and *Office Automation* (Morgall and Vedel, 1985). The Scandinavian PD tradition did not only establish new ways to involve workers in IT design projects, it also helped establish new Data Protection Acts and influenced the Worker Protection and

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[1] We use the term Scandinavian Participatory Design (PD) to refer to the early years of the Participatory Design tradition in the Scandinavian countries and Participatory Design (PD) to refer to this design tradition in general. Several PD researchers cited in this chapter use or have used different terms to refer to the early years of the Participatory Design tradition, such as User-centred Systems Design, Systems Development, and Cooperative Design.
[2] 1975-1980.

Working Environment Acts in Norway, Denmark and Sweden. In addition, these projects brought also the psycho-social issues of the working life to the foreground (Bratteteig, 2004) and new notions were invented to adapt the specialist language of the researchers to the local and situated practices (Nygaard and Bergo, 1975).

Having a Say, the title of Kensing and Greenbaums's (2012) chapter in *Routledge International Handbook of Participatory Design* (Simonsen and Robertson, 2012), reflects the main focus in the early days of PD. It was essential to engage those who were going to be affected by the development and implementation of an IT system in decisions about the design of the system: workers were having decision-making power in the design of new technologies that would affect their work and their skills.

Kensing and Greenbaum consider political economy, democracy, and feminism as the theoretical roots of the early Scandinavian PD tradition. Participatory Design evolved in a time in which workers and activists organised themselves to demand improved working conditions. Their organisations challenged the asymmetrical power relations between trade unions and employer organisations/management and demanded the inclusion of the voices of those in the margins in order to increase their influence at the societal level. The third root of Participatory Design, feminism, helped focus on work dominated by women and to include their voices and skills in the design process. For example, the Florence project (1984-1987) at the University of Oslo, involved nurses – a professional group dominated by women – in the design of an IT system (Bratteteig, 2004).

Bratteteig et al. (2012) describe several design approaches found in Participatory Design. Our own PD practices, in which we focus on the design of technologies and services, are mainly informed by the Use-oriented design approach, which is based on a six-phase iterative design cycle (see Figure 1). In this approach, process and product are of equal importance. The design process enables the emergence of values and definitions of use, while the artefact (product or service), in its different stages of development, enables the exploration of those different definitions of use (Redström, 2008).

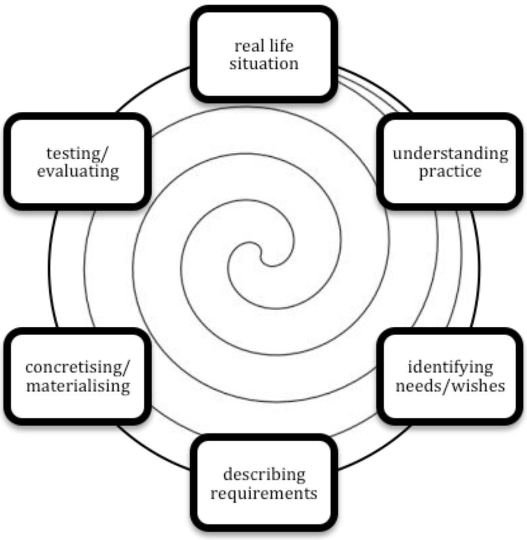

Figure 1. Use-oriented design cycle (based on Bratteteig et al., 2012, p. 128)

As we will discuss below, the values emerging during this design process are materialised in the designed object. The designers and co-designers take design decisions that implicitly and explicitly inscribe values in the final product. Together they present the design's meaning and prescription or *script* for use (Verbeek, 2006). The importance of this process lies in the fact that technology mediates the behaviour of people. The script of a designed product promotes certain use(s) and inhibits other use(s) (ibid.). When there is a discrepancy between the design context and use context, the script will not be strong or stable enough to mediate the behaviour of the users in the way it was envisioned (Latour, 1991). Participatory Design can offer a democratic way to "moralize technology" (Verbeek, 2011). At the same time, we agree with Ihde (1999) and Albrechtslund (2007) that technology is multi-stable, that is, it can have different stable meanings in different use contexts. Participatory Design does not promise a direct connection between design and use, somehow forcing a design's meaning on all use contexts. On the other hand, involving future users as co-designers in the design process significantly increases the chance that the product represents the values and meaning of the future users.

2. Participatory Design's guiding principles

Participation and democracy are the principle values of PD (Bratteteig et al., 2012; Robertson and Wagner, 2012). These values challenge traditional systems design approaches, which are based on a distance between designers and prospective users of the projected information technologies. Having a say in the design process, in all its activities and decisions, enables other principles, such as a design practice based on *equalising power relations* (Kensing and Greenbaum, 2012). In addition, the design process involves co-realisation with a range of participants with their diversity of experiences and knowledge (Bratteteig et al., 2012). The commitment to *democratic practices* results in involving all those who will be affected by the new technology in the design process (Kensing and Greenbaum, 2012). These democratic practices are to be maintained throughout all design activities, enabling trust among all those involved and facilitating a learning process and a commitment to taking responsibility for each other and for the design result. Besides *equalising power* and *democratic practices*, Kensing and Greenbaum mention four other guiding principles: *situation-based actions*, *mutual learning*, *tools and techniques*, and *alternative visions about technology*.

a) Situation-based action

Situation-based action pays attention to people's expertise of their day-to-day activities in work or other practices. In doings and actions, individual or collectively with others people and technology, skills and knowledge are shared and gained. Thus, design is always performed *somewhere* by humans and nonhumans; their activities do not take place in isolation, but are embodied and situated (Suchman, 2002; 2007). People's skills and expertise are implicated in both visible and invisible work (Elovaara et al., 2006; Karasti, 2003; Star and Strauss, 1999). To locate the design activities in people's daily work or other doings may avoid understanding work only as "organizational processes", and approaches work as being "part of the fabric of meanings within and out of which working practices – our own and others' – are made" (Suchman, 1995, p. 58).

b) Mutual learning

Democracy and participation also enable *mutual learning*, learning from each other. Co-designers – such as workers – learn from designers about design and related technological issues. In turn, designers learn about the workplace – the use context, and the workers' activities and skills. In mutual learning, the participants not only share their practical knowledge, competences, and values, "they also learn more about their work themselves" (Karasti, 2001, p. 236). Karasti came to this conclusion through Participatory Design workshops held at the radiology department at the Oulu University Hospital, Finland. The workshops provided the participants the opportunity to scrutinize their own work practices. She continues: "The analytic distance allowed them to articulate meanings of work and to discover previously invisible taken-for-granted aspects of routine practices" (ibid.). Jansson (2007) confirms Karasti's findings: embodied knowledge is implicated in people's day-to-day work and becomes visible in participatory activities.

Although the underlying value, to learn from each other, is implicated in the notion of mutual learning, there is a risk of falling back into dualities and thereby getting caught up in keeping apart the designers and other participating practitioners in the design process. In the dominant discourse in participatory methodologies, there are tendencies to mainly focus on the users and the use context without paying attention to the designers' values and norms and how they are brought into the design setting (Markussen, 1996). In addition, the taken-for-granted-ness of professional designers' expertise has also been questioned (see Bath, 2009; Vehviläinen, 1997). Researchers have used the notion of *design from nowhere* to refer to the disembodied or even invisible professional designer, who is located everywhere (Markussen, 1996; Suchman, 2002). In *Design from somewhere,* Suchman (2002) takes the reverse position: knowledge, including professional expertise, is understood as partial and situated knowledge (Haraway, 1988), dependent on a practice, its history, the involved participants' views and knowledge, and their participation.

Today, Participatory Design is a methodology used in several other disciplines, such as urban planning, architecture, and sustainable development. It also moved from local to global settings, while its methodology has also been challenged by changes in societies and the development of new technologies since the early days of PD, such as transformations in the socio-economic make-up of societies; development of personal technologies, the diffusion of information technology to every aspect of everyday life; and most importantly, more knowledgeable co-designers. For example, the Health Information Systems Programme is a Participatory Design project developing free and open source District Health Information System (DHIS) software. The project started in South Africa in the early 1990s, but has since developed in a global network of participating countries and institutions. The meaning of participation, and the methods to support participation, have changed significantly, from design workshops involving public health researchers and activists with a background in the anti-apartheid struggle and informatics researchers, to a global virtual team of developers in a South-South-North network (Braa and Sahay, 2012).

Ubiquitous computing, smartphones, and other technologies are now available everywhere, at home, at work, in public spaces, challenging the dominant expertise discourse, which in turn challenges Participatory Design. For example, when the Scandinavian PD tradition started, the knowledge of information technology was something only technical experts possessed (Markussen, 1996). These days, co-designers can be expert users in the technology under design, such tech-savvy teenagers in a social media and mobile applications project (e.g., van der Velden and Machniak, 2014) or bioinformaticians in a participatory programming project (Letondal and Mackay, 2004)

c) Tools and techniques

One way to practice participation is by using participatory methods, or *tools and techniques,* which forms another important principle in PD (Kensing and Greenbaum, 2012). A range of methods have been developed and introduced to facilitate participation and cooperative design, such as future workshops, mock-ups, storyboards, scenarios, probes, walk-throughs, games, workshops, cartographic mappings, collaborative prototyping, etc. (Bødker et al., 2004; Brandt et al., 2012; Bratteteig et al., 2012; Mörtberg et al., 2010). Ethnography is also widely used in PD, particularly in the initial phases, to capture the richness of work and other practices (Blomberg and Karasti, 2012). The ethnographer may also work as a facilitator to create dialogues and to enable collaboration (Blomberg et al., 1993). In these kinds of collaborations, everyone contributes with their knowledge and perspectives in participatory dialogues (Christiansen, 2014; Luck, 2003), with the aim to bridge the distance between the various practitioners and to enable questions about "the terms 'we' and 'others'" (Suchman, 1995, p. 61). All PD's methods try to encourage participatory dialogues and to integrate ethical values in the entire design process. In Section 4 we will discuss four of such participatory methods.

Methods, or tools and techniques, play a central role in the creation of an inclusive and democratic design space: "A major strength of Participatory Design is that there is a robust connection between ethical practice and the choice of methods, tools, and techniques" (Robertson and Wagner, 2012, p. 78). Participative methods are a prerequisite to enable people to participate in the design process as experts of their day-to-day work or daily life. Telier et al. (2011) argue this is central to PD: "[In fact, as we shall see,] the origination of participatory design as a design approach is not primarily designers engaging in use, but people (as collectives) engaging designers in their practice" (p. 162).

d) Alternative visions about technology

The early PD projects made clear that *having a voice* in a design process was not enough. *Having a say* in the technology design was necessary for real change to take place: "It appeared to be necessary to create alternative technologies as well as to fight vendors' monopoly over technological choices" (Kensing and Greenbaum, 2012, p. 29). Developing an alternative vision of technology was for instance central to the UTOPIA project, which took place in Sweden from 1981 to 1984. For this purpose a lab-like design space was created in which workers and researchers could jointly experiment with scenarios, paper mock-ups and different technological solutions. Developing alternative visions about technology is still a crucial aspect of PD. It often involves a design process in which an existing technological solution is redesigned or replaced with an alternative solution based on the values of the users and not those of management, vendor or owner.

The guiding principles of Participatory Design are the result of experiences and practices in the early PD projects, which were politically motivated. The guiding principles give Participatory Design also an ethical orientation, as they focus on participation, inclusion, equality, and sharing. In the following section we will discuss this orientation with a focus on values.

3. A value-centred design approach

In their seminal paper "User Participation and Democracy: A Discussion of Scandinavian Research on System Development", Bjerknes and Bratteteig (1995) argue that there has been a turn from politics to ethics in research on user participation in Participatory Design. Focusing on Scandinavian projects supporting *workplace* democracy, the authors differentiate between a political and ethical road to democracy. In the 1970s, system development projects had an explicit political character. Their mandate included pursuing change in laws and agreements that regulated the use of computers in the workplace. The role of the system developer was that of an emancipator. This changed, according to the authors, in the second half of the 1980s. The system developer as an emancipator in a collective political process became the system developer as a professional facilitator of a system development process. The ethical responsibility of the system developer is now based on his/her own individual ethics, which might or might not be supportive of a larger political programme.

Although Bjerknes and Bratteteig argued in their paper for a reintroduction of the collective political dimension in system development, in order to contribute to workplace democracy and workers' rights, the *ethical road* became the hallmark of Participatory Design approach. Participatory Design was presented as a more ethical approach to the design of information systems and other technologies (Robertson and Wagner, 2012). Steen (2013) refers to this shift from politics to ethics as the *ethical turn* in Participatory Design.

3.1 Values and ethical motivation

The discussion of whether politics or ethics, or both, hallmark Participatory Design is partly inspired by discussions on the definitions of and relation between politics and ethics. We take a pragmatic perspective and state that Participatory Design is a value-centred approach to design, since both politics and ethics are value-centred theories. That is, both politics and ethics want to realise values, "the idealized qualities or conditions in the world that people find good" (Brey, 2010, p. 46).

Secondly, Participatory Design is a value-centred design approach because of its *ethical motivation*, which is built on values. Supporting and increasing democratic practices still is an ethical motivation for Participatory Design (Bjerknes and Bratteteig, 1995; Ehn, 1988; Robertson and Wagner, 2012). The latter define PD's ethical motivation as "to support and enhance how people can engage with others in shaping their world, including their workplaces, over time […] working together to shape a better future." (ibid., p. 65). Robertson and Simonsen (2012, p. 5) define this ethical stand of Participatory Design as the recognition of "an accountability of design to the worlds it creates and the lives of those who inhabit them". These descriptions reflect the way in which Participatory Design has broaden its field, moving out from Scandinavian system development focusing on workers rights and workplace democracy to a more encompassing general and global notion of accountability and *shaping a better future*.

3.2 Emerging and dynamic values

Participatory Design projects do not "frontload" (van den Hoven, 2007) a fixed set of values, but they may frontload a set of moral values appropriate to the particular context. This can be explained by PD's central focus on participation, which results in an approach in which the design process is as important as the end result of this process, the designed. All participants in the design process *have a say* in which moral values should inform the design process and the designed. In other words, moral values may also emerge and be co-created during the design process. Participatory Design's tools and techniques support such emergence of values as well as help dealing with conflicting values (see Iversen et al., 2012).

Values also can stay implicit or latent in the design process. Halloran et al. (2009) give the example of three cases in which values only emerged after the participants were challenged by others or by particular developments in the design process. Most importantly, they found that the relationship of values to design is dynamic (ibid):

> [V]alues emerge during co-design work whether or not we look for them. In addition, there is value evolution, values can change and even conflict as the design process unfolds. This bottom-up, data-driven approach to value identification can provide leverage in solving a number of practical co-design problems as the process unfolds; as well as focusing design activity relevant to the users, it can help with the alignment of values between researchers and users, supporting the design relationship, helping users to understand and contribute at functional and technical levels, lead to user insight about their own values and enable the expression of values both during the design process and, ultimately, in the designed artefact (p. 271).

3.3 Whose values

Value Sensitive Design (VSD) is another well-known approach to design for values. The aim of VSD is to advance moral values in design, in particular human welfare; ownership and property; privacy, freedom from bias; universal usability; trust; autonomy; informed consent; accountability; identity; calmness; and environmental sustainability (Friedman and Kahn, 2003). Manders-Huits and Zimmer (2009) see an important difference between VSD and Participatory Design. They acknowledge Participatory Design as a value-centred approach, but argue that PD is "falling short of directly addressing values of moral import, such as privacy or autonomy" (p. 58). They differentiate between design frameworks that "seek to broaden the criteria for judging the quality of technological systems to include the advancement of moral and human values", such as Value Sensitive Design, and design frameworks such as Participatory Design, that promote functional or instrumental values (see also Manders-Huits and van den Hoven, 2009). Their evaluation raises an important question: *whose values* are advanced in these value-oriented design perspectives?

The question of *whose values* was explored early on in the development of the Participatory Design approach (Wagner, 1993, p. 100):

> This raises the larger issue of how "egalitarian values" – equality, inclusivity, sharing, participation – be implemented. […] In question is also the role of systems designers' own values in a PD project. Some may have a tendency to view themselves as impartial deliverers of technical support to organizational "solutions" worked out by users. Others may argue that systems designers' values inevitably enter the process of negotiating conflict over ethical questions.

Values have also become a site of cultural and generational conflicts. In multicultural settings, the principle values of Participatory Design, participation and democracy, need to be explored and understood in the local context (Elovaara et al., 2006; Puri et al., 2004). For example, an on-going Participatory Design project in Namibia shows that participation is not necessarily associated with democracy (Winschiers-Theophilus et al., 2012): "In a hierarchical society lower ranking members are not expected to publicly and openly express opinions, although they are not formally prohibited from doing so. This might seem unjust and counterproductive to participation, when participation is associated with egalitarianism or democracy" (p. 165). A PD project in Cambodia, designing a device that would enable children using a prosthetic leg to walk in the mud, showed that cultural and socio-economic structures prevented a participatory process involving all stakeholders: "the users were raised in a culture where children are not encouraged to express their own opinions but to be obedient towards adults" (Hussain et al., 2012). The designers were able to solve the problem by organising separate design workshops for adults and for children.

3.4 Value practices in Participatory Design

Values play a central role in Participatory Design. The principle values of participation and democracy are perceived as PD's central values as they inform PD's approach and methods. Methods used during the earlier phases in the design process (see Figure 1) enable the emergence of the needs and values of the co-designers. In Scandinavian PD projects of the 1970s and 1980s, the values of worker participation, workplace democracy, together with quality of working life, were also considered the central values informing the designed object (Kensing and Greenbaum, 2012). Iversen et al. (2012) argue against privileging values, including those values associated with PD, such as "participation, democracy, and human welfare" (p. 90). Their design approach focuses on an appreciative judgement of values by the designer through a dialogical process of the emergence, development, and grounding of values. This dialogical process is also used to overcome value conflicts. Appropriate methods are brought in to help the different co-designers to re-engage with their values: "The idea is to create opportunities for them to question and to renegotiate their values, potentially unfinalising their original perceptions of their values. Sometimes, this could even lead to new conceptualisations of their values" (p. 96).

In our own design practices we often use a combination of value practices. We facilitate a process in which values can emerge, but we often also need to *frontload* certain values (van den Hoven, 2007) when they are part of the design brief of a design project. For example, in our design practices in the health care sector, autonomy and privacy are central moral values, but we experienced different understandings of these values between young co-designers and ourselves, especially in an online context. In the same practices, autonomy and privacy are also instrumental values, as patient autonomy and personal health information are regulated by laws and regulations and are enforced through Research Ethics Board reviews and informed consent forms. Sometimes these different meanings of the same values can be in conflict. What autonomy and privacy mean, and how they become materialised in a design, emerges in and through the design process. We thus understand the design process as a *contact zone* (Haraway, 2003; Pratt, 1998; van der Velden, 2010), in which different meanings and understandings of autonomy and privacy *meet* and *grapple* with each other. Such meanings and understandings do not meet as wholes; they are relational entities that enter new relations in the design process. The notion of contact zone helps us to understand that the design process is a space for "communication across irreducible differences" and "situated partial connections" (Haraway, 2003, p. 49). We may never fully know each other's values, but we can meet *respectfully* in our design activities. The design process thus becomes a space for a pluralism of values and for an *agonistic, non-coercive consensus* (Mouffe, 1993).

The dialogical process described by Iversen et al. (2012), which is based on the analysis of three PD projects, is a good example of the PD process as a contact zone. In a dialogical process, the participants do not take non-negotiable positions. Through discussions, observations, visualisations, and interpretations, participants were able to *meet* and *grapple* with other positions, which resulted in the re-negotiation of their own positions. For example, in the case of the design of an interactive school project (ibid., p. 96), the students' values oriented the project towards a student-centred model, questioning the central role of the teachers. The designers introduced fictional inquiry (Dindler and Iversen, 2007), similar to *future workshops* (see Section 4) to facilitate a fictional space in which both students and teachers could play their roles without being threatened. The result was that the teachers began to explore their role as game masters, thus reconceptualising the value of what it means to be a teacher and educator.

Paying attention to multiple voices is foundational in Participatory Design, but this can result in value conflicts. A conflict can become an important resource in the design process (Gregory, 2003), but can also be the result of larger organisational conflicts, which may be *un-dissolvable* (Bødker, 1996). When the value conflict is located in the group of participants itself, PD's wide variety of tools and techniques can be used to explore the different values. Rather than using tools to build consensus, tools and techniques are used to explore this value pluralism, while postponing decisions on the formulation of problems and solutions (Bratteteig and Wagner, 2014, pp. 22–23). Also the Aristotelian concept of *phronesis* has guided dealing with value conflicts. As Ehn and Badham (2002) write:

> "In phronesis, wisdom and artistry as well as art and politics are one. Phronesis concerns the competence to know how to exercise judgement in particular cases. It is oriented towards analysis of values and interest in practice, based on a practical value rationality, which is pragmatic, and context dependent. Phronesis is experience-based ethics oriented towards action" (p. 6)

Flyvbjerg describes *phronetic research* as research with a focus on power and values and the task of the phronetic researcher "to provide concrete examples and detailed narratives of how power works and with what consequences, and to suggest how power might be changed and work with other consequences" (Flyvbjerg, 2001, p. 140). Phronesis can be understood as one of the ethical roots of Participatory Design, enabling imagination and emotions to have a part in the design process (Bratteteig and Stolterman, 1997) and to guide the process "to serve the common good and avoid harming people's possibilities to develop a life of their own" (Kanstrup and Christiansen, 2006, p. 328).

4. Participation, methods, and values

Methods are central to creating an inclusive and democratic design process: they help define and facilitate participant and participation, they enable the expression and exploration of values, and they make use-before-use possible (as will be explained below). We started this chapter with a brief recollection of Participatory Design of the 1970s, which illustrated how participation and its related values became the principal elements in Participatory Design. "[G]enuine participation" is considered both a political and ethical value and the core of Participatory Design (Bødker et al., 2004). In this context, Robertson and Simonsen speak of the "fundamental transcendence of the user's role from being merely an informant to being a legitimate and acknowledged participant in the design process" (2013, p. 5). But what is *genuine* participation? In *Disentangling participation: Power and decision-making in Participatory Design*, Bratteteig and Wagner (2014) explore what they call the most difficult part of Participatory Design: "the sharing of power inherent in the PD approach: in order to collaborate with users as co-designers, the designers need to share their power with them and acknowledge their different and equally valuable expertise" (p. 6). With the spread of Participatory Design to other areas and sectors, participation sometimes became an instrumental value, no longer based on sharing decision-making power with participants, and exploitative (Keinonen, 2010; Shapiro, 2010). Genuine participation requires participatory designers to be mindful about the issue and workings of power.

One important thread in the focus on participation has been the relation of the designers and researchers to the other participants in the design process. In early Participatory Design, they were workers, but their role in the design process was not perceived as part of their position in their workplace. Their role was defined as (future) users of the system under design. This was reflected in the early Scandinavian Participatory Design literature, which uses the term user participation, not worker participation. Greenbaum and Kyng (1991) speak of the *awkwardness* of calling the people, who use computers, users, in the context of a participatory design process, but continue using the term nevertheless. This issue continued to occupy PD researchers. Bødker (1996) wrote: "[…] the term user may be a mistake. However, for lack of a term that covers all the different kinds of workers […] I will utilize the term in this article" (p. 217). Redström (2006; 2008) argues that calling people users, while the things are not yet designed, obscures the process of becoming a user. We therefore prefer to use the term *co-designers* for the people with whom and for whom we design.

4.1 Multiple voices and silences

The focus in a technology design process is often on what seems the most obvious solution to a problem: the design a technology that can do something more efficient, effective, smarter or entertaining. In Participatory Design, we try to postpone *the most obvious solution* in order to enable co-designers to explore alternative visions of technology and their different uses and effects. This is facilitated by Participatory Design's commitment to democratic practices, which enables people who otherwise might be invisible or marginalised to have a voice in the design process. Elovaara, Igira and Mörtberg (2006, p. 113) present the following notes from a research diary of a conversation in a design process:

> "We are laughing and talking. One of our participants is telling a wonderful story from her everyday work. One citizen visited her. The woman was behind with the payments of day nursery. She wanted to pay in cash but the system did not accept any real money. So the civil servant said to the person: "Let's go together to the bank. Then I'll see to it that you fix it and I get a receipt. When I go back to the office I register your payment. After that everything is in order". And that's what they did: they marched together to the nearby bank. Everything worked out smoothly. And everybody was happy: the citizen (she could pay the fee for day nursery and not end up in the enforcement register), the civil servant (she could do her job and receive the expected fee) and the municipality (they got their money). Then Pirjo [one of the design researchers] asked if citizens couldn't use the municipality web site to check and pay their taxes and other fees – develop a self-service municipality? We started to imagine and make scenarios. And suddenly I heard the civil servant who sat next to me, she said, very, very quietly: "But then Anna (referring to the civil servant working with the municipal fees, sorting them, posting the payment forms, reminding etc.), will become unemployed?!.

The risk to be replaced by technology was the value that emerged from the participative methods in the project. Elovaara et al. (ibid.) refer to this as an emergent ethical issue related to knowledge and visibility. In order to *do no harm* to the co-designers, they stress the need to "take the PD-core ideas seriously" and to find alternative visions to what was the most obvious solution, that is, to replace a person with technology.

To pay attention to multiple voices is central in PD, including voices that are silent or are silenced. Stuedahl (2004) discusses silences in a Norwegian Participatory Design project called NEMLIG (Net- and multimedia based learning environments – 2000-2002). The project was conducted at a middle-sized Norwegian graphical company and the purpose was to design net-based features for learning at work. The participants involved in the projects were designers, researchers, graphical workers, typographers, and graphical designers. The pilot was managed by a union-based competence centre, which wasn't well versed in participatory design methods. The competence centre organised a range of design meetings. The third design meeting resulted in a breakdown when the graphical workers stopped the design process. Mörtberg and Stuedahl (2005, p. 142) write:

> This is what happened during a workshop session, when the systems developer explained possible solutions for the system: the users turned silent. They did not ask questions, and they did not comment on the ideas the system

> developers presented. After one hour of silence – meaning users' silence, while the system developers discussed a variety of issues – the head of the user group spoke up and asked the system developers [for a break].

The break had the effect of a breakdown (Luck, 2007). Afterwards the co-designers were able to express their grievances and to bring their values and needs into the design process.

Silences may emerge during a design process and it has implications for the participants' actions. The challenge is how to respect silences and to create a democratic design space, which gives voice to all, included to those who are silent. Sensitivity, as a virtue, is necessary to pay attention to silences (Karasti, 2001; Mörtberg and Stuedahl, 2005), as there is a tendency to neglect silences (Finken and Stuedahl, 2008).

The two examples of our research diaries show that a participatory design process is not only about designing an object. The designers and co-designers work together in the design process, making visible and concretising their knowledge, values, and needs into the designed. Simultaneously, understandings of the use(s) and users of the design emerge in this iterative process. For example, Anna, in the first example, emerges as a non-user because she would loose her job if the municipality would start using an online payment system. The co-designers in the second example first emerge as non-users and express this through their silence. After the breakdown in the process, the co-designers emerge as potential users, depending on how the designers would address their needs. The central design challenge in both cases is that the *use* of the designed object can only be envisioned or anticipated, because there is no actual artefact that can be used: this is what is called *use-before-use* (Ehn, 2008; Redström, 2006). Use-before-use requires participatory methods that engage the co-designers in telling, making, and enacting use (Brandt et al., 2012).

4.2 Values and 'designing for use-before-use'

Participatory Designer Pelle Ehn has described PD as "designing for use-before-use" (2008). Over the years, a range of methods have been designed, developed, and evaluated that engage and support co-designers in envisioning use of the designed. These methods aim to express and explore values as well as to inscribe these values in the product of the design process, the materialisation of values (Verbeek, 2006). Although all participatory design methods and activities elicit information, discussion, reflection, and learning, some methods are particularly suited for expressing, exploring, and materialising values by engaging co-designers in telling, making, and enacting use.

Certain PD methods are particularly productive for exploration and engagement in the early stages of the design process, when the focus is on *understanding practices* and *identifying needs and wishes* (see Figure 1). During a Participatory Design process, values are expressed and explored and then become materialised in the form of an object. The materialisation of values is the result of interactions between the designers and co-designers and material objects (materials, tools, mock-ups, prototypes, etc.). In this process, co-designers become users and the material object becomes a product or a service. "Technology is society made durable," wrote Bruno Latour (1991). In a participatory design process, ideas of what is good (values), useful (functions), and beautiful (aesthetics) are made durable in inscriptions in the design, such as choices in materials, options for use, colour use, flexibility, etc. Anthropological and sociological studies of the design and use of technology have shown that these inscriptions of values in a design are not prescriptive. Although they can function as a *script* for use (Akrich, 1992), if and how an artefact will be used, is based on the strength of the inscriptions (Latour, 1991). On the other hand, the fact that future users of an artefact have been involved in the inscription process often results in artefacts that matter to them and increases the likelihood that they will use it the way it was envisioned.

4.3 Methods for exploring, engaging with, and materialising values

There is a large body of design methods in use in Participatory Design (e.g., Brandt et al., 2012; Sanders, Brandt and Binder, 2010; Spinuzzi, 2005; Wölfel and Merritt, 2013). We will discuss four of these methods: card methods, mapping methods, future workshop, and participatory prototyping (see Table 1).

Table 1. Participatory methods for design for values

Card methods	Card methods in PD are used to inspire and explore and express emotions and values. They often come with specific instructions on how to use them to engage with fellow co-designers and to facilitate communication with the designers. For an overview of card methods, see Wölfel and Merritt (2013)
Mapping methods	Mapping methods are used to holistically map, express, and explore local knowledge; and to enable those who will be affected by the design to express their values and to be actively involved in the design of an artefact.
Future workshop	A method for planning and enacting possible futures, through the involvement of various participants (stakeholders) and with the use of various phases, towards a joint proposal for change (Jungk and Müllert, 1981).
Participatory prototyping	Participatory prototyping is the process of generating, evaluating, and concretising design ideas with active involvement of the future users (Lim, Stolterman andTenenberg, 2008)

a) Card methods

Cards and card sorting methods are widely used in PD. Cards are rectangular or square pieces of paper or carton (approximately the size of playing cards) containing text, an image or both. They have a large use potential because of their tangibility, ease of use, and inexpensive production. Cards are used for different purposes such as idea generation, inspiration, engagement, empathy, and to overcome problems that appear in a design process (Wölfel and Merritt, 2013). In PD, cards are often customisable: co-designers can add new cards or change cards. Cards can also be created during a discussion, expressing the themes or issues, which then can be sorted on priority or concept. Card sorting is a widely used design method in which the co-designer(s) organise the cards into categories or select particular cards to visualise processes, express priorities or inspire creative processes.

Figure 2. Inspiration Cards (Photo: KULU)

We designed and used a set of *Inspiration Cards* (Halskov and Dalsgård, 2006) for our design work with teenagers with chronic health challenges preparing for the transition to the adult hospital. The cards addressed four categories connected with the transition process: *important people*, *things*, *feelings*, and *skills*. The cards consisted of an image with a one- or two-word concept description. The teen co-designers selected the cards they deemed important in each of the three stages in the transition process: preparing for transition, transitioning, and after transition. We also provided them with ‘empty’ cards – cards without images or texts – providing them the opportunity to make their own inspiration cards and to add them to the stock of cards. In the process of selecting and organising the cards and explaining their choices, the teens engaged with what was important to them. They visualised and verbalised what was important for them to know, to do, to feel or to experience in the context of the transition process. The method facilitated the emergence of values in a more spontaneous and comprehensive manner and enabled better insight in the values at stake for themselves and for the designers. The Inspiration Cards are used to explore and engage with values early in the Use-centred design cycle (see Figure 1).

b) Mapping methods

Mapping methods are used to express and explore local knowledge. The resulting maps give holistic visualisations of geographies, contexts, life histories, workflows, or connections and relationships of the co-designers (Lanzara and Mathiassen, 1985). Examples are context mapping (Visser et al., 2005) and cognitive mapping (Goodier and Soetanto, 2013).

The *Cartographic Mapping method* is a method for making peoples’ doings and activities in their everyday and working lives visible (Elovaara and Mörtberg, 2010). The method is collective, simple, cheap, and the material is familiar to all participants. The participants do not need to make any preparations in advance. The simplicity of the method encourages participants to start telling their stories and visualise their work or activities in everyday life and working life. The facilitators of the mapping activity prepare the workshop by collecting images of people and things and other materials such as yarn, pencils, coloured paper, post-it notes, and large sheets to paste the images on. The workshop starts with a presentation of the method and the chosen topic. The room is arranged like a design space with a variety of materials. The participants are asked to take one sheet and to choose an image that presents them. The participants then start to include images of people, things and of their own activities (see Figure 3). During the mapping activities the facilitators ask the participants to clarify their choice of images and the connections between people and things pasted on the sheets. After the maps have been created, the participants are asked to share their stories with each other. The mapping activity initiates and contributes to a process in which values become expressed and materialised through the visualisations, the informal interviews, and the participants' oral stories of their cartographies. These stories can then be translated with the use of other methods and techniques, such as storyboards or mock-ups.

Figure 3. Cartographic Mapping

The method was developed and used in our research project *From government to e-government: skills, gender, technology, and learning* (Elovaara et al., 2006; Elovaara and Mörtberg, 2010; Mörtberg et al., 2010). The purpose was to examine how the civil servants' skills and knowledge could be integrated in the design of the so-called e-service society. The method has also been used in other design projects, such as projects with an aim to design with and for elderly in their everyday lives.

c) Future workshop

The *future workshop* method was originally developed by Robert Jungk and Norbert Müllert (1981) to involve citizens in public decision-making processes, such as city planning. The method's basic principle, to enable participants to have a say, made it also of interest to PD (Kensing and Madsen, 1991). The methods phases are: preparation, critique, fantasy, realisation, and follow-up. In the *preparation* phase the room is designed to create a welcoming atmosphere and the method, the theme and the facilitator are introduced. *Critique* of the current situation is generated in the next phase. Post-it notes can be used to write down keywords and these can then be pasted on a wall to make them visible for all. When a rich image of the current situation is generated, the keywords are organised into categories followed by a prioritisation. In the *fantasy* phase ideas are generated – a brainstorming without restrictions – and the critique is turned into something positive. Post-it notes can be used in a similar way as in the critique phase. Before the participants move to the *realisation* phase, the generated ideas are analysed and prioritised. In the realisation phase, the participants review their visions and ideas and discuss the possibility for implementation. This is also a collaborative activity with the aim to create a joint proposal – an action plan, for how to change the current situation. A *follow-up* activity is also included in the method.

Future workshops were used in one of our projects with the aim to implement an automatic planning system in a handheld computer in a homecare practice (Jansson, 2007; Jansson and Mörtberg, 2011). A range of occupational groups involved in home care of elderly were invited to participate in the future workshops, in order to enable all stakeholders to be actively involved and to define their demands in the design and implementation of a digital artefact and new work practices. Through the involvement of various stakeholders, multiple values and views emerged out of the activities. This became visible during the generation of critique and visions, the systematisation of the suggestions, their prioritisation, and finally also when the participants consider possibilities for realisation. Future workshops for technology design produce action plans as outcome, which form the first step in the materialisation of requirements in the Use-centred design cycle (see Figure 1).

d) Participatory prototyping

Participatory prototyping, also called cooperative prototyping (Bødker and Grønbæk, 1991) or collaborative prototyping, is one of the most important methods used in PD. Participatory prototyping is the activity in the design process in which values are translated into requirements and become materialised in a designed object. Prototypes can differ in material (paper, digital), resolution (mock-ups, low- and high-fidelity), and scope: they can be used to focus on a particular dimension of a design idea, which enables a more in-depth exploring (Lim et al., 2008).

In Participatory Design, prototyping creates a shared design space for designers and co-designers in which tensions between 'what is' and 'what could be' are explored through enacting scenarios, such as use situations (use-before-use). Prototyping can also provoke hindrances and new possibilities (Brodersen et al., 2008) or opportunities and dilemmas (Hillgren et al., 2011). Prototyping is an iterative process, often evolving from a low-fidelity prototype to a high-fidelity prototype with all the specifications of the finished product.

Several prototyping techniques can be used in one design process. In the KULU project, a design project with teenagers with chronic health challenges, we used sketches, paper mock-ups, as well as digital prototypes in the design of patient social media and mobile applications (KULU, 2014). Our concern with the digital prototypes was that they might look too 'finished' and thus present a solution before the design problem was fully explored. On the other hand, the digital prototypes showed the teens a concrete rendering of their values and wishes, which encouraged them to engage with the prototypes, thus continuing the design process.

Figure 4. Calendar function
(Aasen, 2014)

Privacy is one of the central values in the design projects with the teenagers. Research on teenagers with chronic health challenges and privacy in Canada has shown that most teens separate their identity as a patient from their (online) identity as a teenager and that this affects their online privacy behaviour (van der Velden and El Emam, 2013; van der Velden and Machniak, 2014). During the prototyping activities with Norwegian teenage patients a similar understanding of privacy emerged and materialised in different design projects. One example was the calendar function in a mobile application (app) for patient self-management. All teenage co-designers used the calendar function on their smart phone, but all wanted to have another calendar function within the new app. In a participatory prototyping session, it became clear that there was no consensus among the teenage co-designers on the use of the calendar function on the app. Some wanted to use it solely for medical reminders and appointments, while others wanted to use it for both medical and non-medical reminders and appointments. Designers and co-designers agreed on a new prototype with colours and audio to differentiate between medical (orange) and non-medical appointments and reminders (green) (Aasen, 2014), and thus provided the co-designers with different use scenarios for future use: medical-only, mixed medical/non-medical, or both medical or non-medical but not mixed (Figure 4 is an example of a mixed medical/non-medical calendar). The participatory prototyping session became a space for engaging with multiple understandings of privacy, which were concretised in a new calendar prototype, which was flexible enough to support the three use-scenarios.

5. Open Issues and Future Work

Genuine participation and power relations are two of the main issues in Participatory Design. There is no blueprint for addressing these challenges, but an on-going critical and reflexive engagement with PD's guiding principles and design practices (e.g., Bratteteig and Wagner, 2014) may facilitate ethical practices. It is here important to remember that not only people, but also the structural arrangement in a design project, such as the allocation of resources or the choice of methods affects the power to decide (ibid.). Reflexive and critical approaches become especially important in design projects with vulnerable groups, such as children, patients, and elderly. Several ethical heuristics for PD projects have been developed, which can be applied before and, iteratively, during a design process (e.g., Lahti et al., 2012; Phelan and Kinsella, 2013; Read et al., 2013; Robertson and Wagner, 2012, p. 82).

Another open issue is the political dimension in PD. Many industry-based projects lack an analysis of the larger socio-economic and cultural structures in which we design, produce, and use artefacts, as the focus is on the production of consumer goods (Bergvall-Kåreborn and Ståhlbrost, 2008). The result is the risk of a more instrumental use of the co-designers in a design project, as Beck already observed in 2002: "A politicised agenda for PD would need to centrally address, then, the legitimacy of anyone not only to propose solutions, but to suggest what the problems are" (Beck, 2002, p. 83). Fry (2009; 2011), concerned with the issue of sustainment, argues that PD could strengthen the voice of the *common good* and de-legitimise *consumer democracy*. One way to address these concerns is to let the future generations *have a say* in today's PD projects (van der Velden, 2014). Bringing a multi-generational perspective into PD may result in frontloading certain values, such as sustainment, and the design of new methods or re-design of existing methods, such as the *future workshop*. Also the insights of other design approaches, such as Design Futuring (Fry, 2009) and Metadesign (Fischer et al., 2004; Wood, 2007) can strengthen PD's political dimension.

Future work in Participatory Design may further explore the relation between values and PD methods. Methods play a central role in creating a space for the emergence of values and in engaging designers and co-designers in the expression and exploration of these values. Overviews of participatory methods are helpful (e.g., Sanders et al., 2010; Wölfel and Merritt, 2013), but focus mainly on functionality and application area of methods. Reviews of participatory methods based on how they may support the exploration, engagement with, and materialisation of methods may provide participatory designers with an important tool for strengthening their value practices.

6. Concluding remarks

Participatory Design's principle values, *democracy* and *participation*, make it an important methodology for Design for Values. These values shape the design process, resulting in: a) a genuine engagement with the people who will be using the outcome of the process; and b) the use of design methods that focus on creating a shared design space in which designer and co-designer values are expressed and become materialised in a product or service.

We have argued that the design process in a PD project is as important as its outcome. During the design process, other goals are accomplished, such as mutual learning, reflection, and skill acquisition, which have a value that is independent of the final outcome of the process. Participatory Design's guiding principles prevent the design process to become a purely instrumental phase for eliciting user input for a final product. By facilitating genuine participation, engaging the experiences, skills, needs, and values of the co-designers, the design process becomes a space in which alternative visions about technology are envisioned and anticipated. Through the creative use of participatory methods, the Participatory Design process enables use-before-use scenarios, in which co-designers can try out or engage with the result(s) before it is produced or implemented. The result of this process is a product or service that matters, because the co-designers' values and needs have become materialised in the design.

Cross-References

Design Methods in Design for Values
Value Sensitive Design
Design for the Values of Presence
Design for the Values of Democracy and Justice

References

Aasen, N. (2014). *Transisjonsapp - Ansvar for egen helse* (Master Thesis). University of Oslo, Oslo.

Akrich, M. (1992). The De-scription of Technical Objects. In W. Bijker & J. Law (Eds.), *Shaping Technology* (pp. 205–224). Cambridge: MIT Press.

Albrechtslund, A. (2007). Ethics and technology design. *Ethics and Information Technology*, *9*(1), 63–72.

Bath, C. (2009). Searching for methodology: Feminist technology design in computer science. *Online Proc. Gender & ICT*. Retrieved from http://www.informatik.uni-bremen.de/soteg/gict2009/proceedings/GICT2009_Bath-geloescht.pdf

Beck, E. (2002). P for Political: Participation is Not Enough. *Scandinavian Journal of Information Systems*, *14*(1).

Bergvall-Kåreborn, B., & Ståhlbrost, A. (2008). Participatory Design: One Step Back or Two Steps Forward? In *Proceedings of the Tenth Anniversary Conference on Participatory Design 2008* (pp. 102–111). Indianapolis, IN, USA: Indiana University.

Bjerknes, G., & Bratteteig, T. (1995). User participation and democracy: A discussion of Scandinavian research on system development. *Scandinavian Journal of Information Systems*, *7*, 73–73.

Blomberg, J., & Karasti, H. (2012). Ethnography: positioning ethnography within Participatory Design. In *Routledge International Handbook of Participatory Design* (pp. 86–116). New York: Routledge.

Blomberg, J. L., Giacomi, J., Mosher, A., & Swenton-Hall, P. (1993). Ethnographic Field Methods and Their Relation to Design. In *Participatory Design: Principles and Practices* (pp. 123–156). Hillsdale, N.J.: Lawrence Erlbaum Associates.

Bødker, K., Kensing, F., & Simonsen, J. (2004). *Participatory IT Design: Designing for Business and Workplace Realities*. MIT Press.

Bødker, S. (1996). Creating Conditions for Participation: Conflicts and Resources in Systems Development. *Hum.-Comput. Interact.*, *11*(3), 215–236.

Bødker, S., & Grønbæk, K. (1991). Cooperative prototyping: users and designers in mutual activity. *International Journal of Man-Machine Studies*, *34*(3), 453–478.

Braa, J., & Sahay, S. (2012). Health Information Systems Programme: Participatory Design within the HISP network. In *Routledge International Handbook of Participatory Design* (pp. 235–256). New York: Routledge.

Brandt, E., Binder, T., & Sanders, E. B.-N. (2012). Tools and techniques: ways to engage telling, making, and enacting. In *Routledge International Handbook of Participatory Design* (pp. 145–181). New York: Routledge.

Bratteteig, T. (2004). *Making Change. Dealing with relations between design and use* (Dr. Philos dissertation). Department of Informatics, University of Oslo.

Bratteteig, T., Bødker, K., Dittrich, Y., Mogensen, P. H., & Simonsen, J. (2012). Methods: organising principles and general guidelines for Participatory Design projects. In *Routledge International Handbook of Participatory Design* (pp. 117–144). New York: Routledge.

Bratteteig, T., & Stolterman, E. (1997). Design in groups—and all that jazz. *Computers and Design in Context*, 289–316.
Bratteteig, T., & Wagner, I. (2014). *Disentangling participation: Power and decision-making in Participatory Design*. Springer.
Brey, P. (2010). Values in technology and disclosive computer ethics. *The Cambridge Handbook of Information and Computer Ethics*, 41–58.
Brodersen, C., Dindler, C., & Iversen, O. S. (2008). Staging imaginative places for participatory prototyping. *CoDesign*, *4*(1), 19–30.
Christiansen, E. (2014). From "ethics of the eye" to "ethics of the hand" by collaborative prototyping. *Journal of Information, Communication and Ethics in Society*, *12*(1), 3–9.
Dindler, C., & Iversen, O. S. (2007). Fictional Inquiry—design collaboration in a shared narrative space. *CoDesign*, *3*(4), 213–234.
Ehn, P. (1988). *Work-oriented Design of Computer Artifacts*. Stockholm: Arbetslivscentrum.
Ehn, P. (2008). Participation in Design Things. In *Proceedings of the Tenth Anniversary Conference on Participatory Design 2008* (pp. 92–101). Indianapolis, IN, USA: Indiana University.
Ehn, P., & Badham, R. (2002). Participatory Design and the Collective Designer. *PDC*, 1–10.
Ehn, P., & Sandberg, Å. (1979). Systems Development: On strategy and ideology. *Data*, (4).
Elovaara, P., Igira, F. T., & Mörtberg, C. (2006). Whose Participation? Whose Knowledge?: Exploring PD in Tanzania-Zanzibar and Sweden. In *Proceedings of the Ninth Conference on Participatory Design: Expanding Boundaries in Design - Volume 1* (pp. 105–114). New York, NY, USA: ACM.
Elovaara, P., & Mörtberg, C. (2010). Carthographic Mappings: Participative Methods. In *Proceedings of the 11th Biennial Participatory Design Conference* (pp. 171–174). New York, NY, USA: ACM.
Finken, S., & Stuedahl, D. (2008). Silence' As an Analytical Category for PD. In *Proceedings of the Tenth Anniversary Conference on Participatory Design 2008* (pp. 170–173). Indianapolis, IN, USA: Indiana University.
Fischer, G., Giaccardi, E., Ye, Y., Sutcliffe, A. G., & Mehandjiev, N. (2004). Meta-design: a manifesto for end-user development. *Commun. ACM*, *47*(9), 33–37. doi:10.1145/1015864.1015884
Flyvbjerg, B. (2001). *Making Social Science Matter: Why Social Inquiry Fails and How it Can Succeed Again*. Cambridge University Press.
Friedman, B., & Kahn, P. H. (2003). The Human-computer Interaction Handbook. In J. A. Jacko & A. Sears (Eds.), (pp. 1177–1201). Hillsdale, NJ, USA: L. Erlbaum Associates Inc.
Fry, T. (2009). *Design Futuring: Sustainability, ethics and new practice*. Oxford: Berg Publishers Ltd : Macmillan Distribution.
Fry, T. (2011). *Design as politics*. New York: Berg.
Goodier, C. I., & Soetanto, R. (2013). Building future scenarios using cognitive mapping. *Journal of Maps*, *9*(2), 203–217.
Greenbaum, J. M., & Kyng, M. (1991). *Design at work: cooperative design of computer systems*. Hillsdale, N.J.: L. Erlbaum Associates.
Gregory, J. (2003). Scandinavian approaches to participatory design. *International Journal of Engineering Education*, *19*(1), 62–74.
Halloran, J., Hornecker, E., Stringer, M., Harris, E., & Fitzpatrick, G. (2009). The value of values: Resourcing co-design of ubiquitous computing. *CoDesign*, *5*(4), 245–273.
Halskov, K., & Dalsgård, P. (2006). Inspiration Card Workshops. In *Proceedings of the 6th Conference on Designing Interactive Systems* (pp. 2–11). New York, NY, USA: ACM.
Haraway, D. (1988). Situated Knowledges: The Science Question in Feminism and the Privilege of Partial Perspective. *Feminist Studies*, *14*(3), 575–599.
Haraway, D. (2003). *The Companion Species Manifesto: Dogs, People, and Significant Otherness*. Chicago: Prickly Paradigm Press.
Hillgren, P.-A., Seravalli, A., & Emilson, A. (2011). Prototyping and infrastructuring in design for social innovation. *CoDesign*, *7*(3-4), 169–183.
Hussain, S., Sanders, E. B.-N., & Steinert, M. (2012). Participatory Design with Marginalized People in Developing Countries: Challenges and Opportunities Experienced in a Field Study in Cambodia. *International Journal of Design*, *6*(2).
Ihde, D. (1999). Technology and prognostic predicaments. *AI & SOCIETY*, *13*(1-2), 44–51.
Iversen, O. S., Halskov, K., & Leong, T. W. (2012). Values-led participatory design. *CoDesign*, *8*(2-3), 87–103.

Jansson, M. (2007, November 15). *Participation, knowledges, and experiences: design of IT- systems in e-home health care* (Doctoral Thesis). Luleå Technical University. Retrieved from http://epubl.ltu.se/1402-1544/2007/56/index-en.html

Jungk, R., & Müllert, N. R. (1981). *Zukunftswerkstätten, Wege zur Wiederbelebung der Demokratie (Future Workshops: Ways to revive democracy)*. Munchen: Goldman.

Kanstrup, A. M., & Christiansen, E. (2006). Selecting and Evoking Innovators: Combining Democracy and Creativity. In *Proceedings of the 4th Nordic Conference on Human-computer Interaction: Changing Roles* (pp. 321–330). New York, NY, USA: ACM.

Karasti, H. (2001). *Increasing Sensitivity towards Everyday Work Practice in System Design*.

Karasti, H. (2003). Gendered Expertise and Participation in Systems Design. In *How to make a difference?: Information Technology, Transnational Democracy and Gender* (pp. 29–49). Luleå: Luleå University of Technology.

Keinonen, T. (2010). Protect and appreciate–Notes on the justification of user-centered design. *International Journal of Design*, *4*(1), 17–27.

Kensing, F., & Greenbaum, J. (2012). Heritage: Having a say. In *Routledge International Handbook of Participatory Design* (pp. 21–36). New York: Routledge.

Kensing, F., & Madsen, K. H. (1991). Generating Visions: Future workshops and metaphorical design. In *Design at Work: Cooerative design of computer systems* (pp. 155–168). Hillsdale, N.J.: Lawrence Erlbaum Associates.

KULU - Kul teknologi for unge med langvarige helseutfordringer. (2014). Retrieved May 30, 2014, from http://www.kulu.no/english.php

Lahti, A., Naraha, S., Svensson, J., & WärneståI, P. (2012). Ethical Heuristics – A Tool for Applying Ethics in User-Involved IS Projects. Presented at the Nordic Contributions in IS Research: Third Scandinavian Conference on Information Systems, SCIS 2012, Sigtuna, Sweden, August 17-20, 2012, Proceedings. Retrieved from http://hh.diva-portal.org/smash/record.jsf?pid=diva2:548656

Lanzara, G. F., & Mathiassen, L. (1985). Mapping situations within a system development project. *Information & Management*, *8*(1), 3–20.

Latour, B. (1991). Technology is society made durable. In J. Law (Ed.), *A sociology of monsters: Essays on power, technology, and monsters* (pp. 103–131). London: Routledge.

Letondal, C., & Mackay, W. E. (2004). Participatory Programming and the Scope of Mutual Responsibility: Balancing Scientific, Design and Software Commitment. In *Proceedings of the Eighth Conference on Participatory Design: Artful Integration: Interweaving Media, Materials and Practices - Volume 1* (pp. 31–41). New York, NY, USA: ACM.

Lim, Y.-K., Stolterman, E., & Tenenberg, J. (2008). The Anatomy of Prototypes: Prototypes As Filters, Prototypes As Manifestations of Design Ideas. *ACM Trans. Comput.-Hum. Interact.*, *15*(2), 7:1–7:27.

Luck, R. (2003). Dialogue in participatory design. *Design Studies*, *24*(6), 523–535.

Luck, R. (2007). Learning to talk to users in participatory design situations. *Design Studies*, *28*(3), 217–242.

Manders-Huits, N., & van den Hoven, J. van den. (2009). The Need for a Value-Sensitive Design of Communication Infrastructures. In P. Sollie & M. Düwell (Eds.), *Evaluating New Technologies* (pp. 51–60). Springer Netherlands.

Manders-Huits, N., & Zimmer, M. (2009). Values and pragmatic action: The challenges of introducing ethical intelligence in technical design communities. *International Review of Information Ethics*, *10*(2), 37–45.

Markussen, R. (1996). Politics of intervention in design: feminist reflections on the Scandinavian tradition. *Ai & Society*, *10*(2), 127–141.

Morgall, J., & Vedel, G. (1985). Office Automation: The Case of Gender and Power. *Economic and Industrial Democracy*, *6*(1), 93–112.

Mörtberg, C., Bratteteig, T., Wagner, I., Stuedahl, D., & Morrison, A. (2010). Methods That Matter in Digital Design Research. In I. Wagner, T. Bratteteig, & D. Stuedahl (Eds.), *Exploring Digital Design* (pp. 105–144). Springer London.

Mörtberg, C., & Stuedahl, D. (2005). Silences and Sensibilities: Increasing Participation in IT Design. In *Proceedings of the 4th Decennial Conference on Critical Computing: Between Sense and Sensibility* (pp. 141–144). New York, NY, USA: ACM.

Mouffe, C. (1993). *The Return of the Political*. London: Verso.

Nygaard, K. (1974). *Planlegging, styring og databehandling: Grunnbok for fagbevegelsen Del 2 Datamaskiner, systemer og språk*. Oslo: Tiden Norsk Forlag.

Nygaard, K., & Bergo, O. T. (1975). The Trade Unions - New users of research. *Personnel Review*, *4*(2), 5–10.
Phelan, S. K., & Kinsella, E. A. (2013). Picture This . . . Safety, Dignity, and Voice—Ethical Research With Children Practical Considerations for the Reflexive Researcher. *Qualitative Inquiry*, *19*(2), 81–90.
Pratt, M. L. (1998). Arts of the Contact Zone. *Negotiating Academic Literacies: Teaching and Learning Across Languages and Cultures*, *91*, 171.
Puri, S. K., Byrne, E., Nhampossa, J. L., & Quraishi, Z. B. (2004). Contextuality of Participation in IS Design: A Developing Country Perspective. In *Proceedings of the Eighth Conference on Participatory Design: Artful Integration: Interweaving Media, Materials and Practices - Volume 1* (pp. 42–52). New York, NY, USA: ACM.
Read, J. C., Horton, M., Sim, G., Gregory, P., Fitton, D., & Cassidy, B. (2013). CHECk: A Tool to Inform and Encourage Ethical Practice in Participatory Design with Children. In *CHI '13 Extended Abstracts on Human Factors in Computing Systems* (pp. 187–192). New York, NY, USA: ACM.
Redström, J. (2006). Towards user design? On the shift from object to user as the subject of design. *Design Studies*, *27*(2), 123–139.
Redström, J. (2008). RE:Definitions of use. *Design Studies*, *29*(4), 410–423.
Robertson, T., & Wagner, I. (2012). Ethics: engagement, representation and politics-in-action. In *Routledge International Handbook of Participatory Design* (pp. 64–85). New York: Routledge.
Sanders, E. B.-N., Brandt, E., & Binder, T. (2010). A framework for organizing the tools and techniques of participatory design. In *Proceedings of the 11th biennial participatory design conference* (pp. 195–198). ACM. Retrieved from http://dl.acm.org/citation.cfm?id=1900476
Shapiro, D. (2010). A Modernised Participatory Design? *Scandinavian Journal of Information Systems*, *22*(1), 69–76.
Simonsen, J., & Robertson, T. (2012). *Routledge international handbook of participatory design*. New York: Routledge.
Spinuzzi, C. (2005). The methodology of participatory design. *Technical Communication*, *52*(2), 163–174.
Star, S. L., & Strauss, A. (1999). Layers of Silence, Arenas of Voice: The Ecology of Visible and Invisible Work. *Computer Supported Cooperative Work (CSCW)*, *8*(1-2), 9–30.
Steen, M. (2013). Virtues in Participatory Design: Cooperation, Curiosity, Creativity, Empowerment and Reflexivity. *Science and Engineering Ethics*, *19*(3), 945–962.
Suchman, L. (1995). Making Work Visible. *Commun. ACM*, *38*(9), 56–ff.
Suchman, L. (2002). Located accountabilities in technology production. *Scandinavian Journal of Information Systems*, *14*(2), 91–106.
Suchman, L. (2007). *Human-Machine Reconfigurations: Plans and Situated Actions (2nd edition)*. Cambridge: Cambridge University Press.
Telier, A., Binder, T., De Michelis, G., Ehn, P., Jaccuci, G., Linde, P., & Wagner, I. (2011). *Design Things*. Cambridge Mass.: MIT Press.
Van den Hoven, J. van den. (2007). ICT and Value Sensitive Design. In P. Goujon, S. Lavelle, P. Duquenoy, K. Kimppa, & V. Laurent (Eds.), *The Information Society: Innovation, Legitimacy, Ethics and Democracy In honor of Professor Jacques Berleur s.j.* (pp. 67–72). Springer US. Retrieved from http://link.springer.com/chapter/10.1007/978-0-387-72381-5_8
Van der Velden, M. (2010). Design for the contact zone. Knowledge management software and the structures of indigenous knowledges. In F. Sudweeks, H. Hrachovec, C. Ess, F. Sudweeks, H. Hrachovec, & C. Ess (Eds.), *Cultural Attitudes towards Technology and Communication 2010 Proceedings of the Seventh International Conference on Cultural Attitudes towards Technology and Communication Vancouver, Canada, 15-18 June 2010* (pp. 1–18). Murdoch: School of Information Technology, Murdoch University. Retrieved from http://sammelpunkt.philo.at:8080/2002/
Van der Velden, M. (2014). Re-politicising Participatory Design: What we can learn from Fairphone. In *Ninth International Conference on Culture, Technology, and Communication*. Oslo. Retrieved from http://philo.at/ocs2/index.php/oslo14/ctnewd14/paper/view/295
Van der Velden, M., & El Emam, K. (2013). "Not all my friends need to know": a qualitative study of teenage patients, privacy, and social media. *Journal of the American Medical Informatics Association: JAMIA*, *20*(1), 16–24.
Van der Velden, M., & Machniak, M. (2014). Colourful Privacy: Designing Visible Privacy Settings with Teenage Hospital Patients (pp. 60–65). Presented at the ACHI 2014, The Seventh

International Conference on Advances in Computer-Human Interactions. Retrieved from http://www.thinkmind.org/index.php?view=article&articleid=achi_2014_3_30_20220
Vehviläinen, M. (1997). *Gender, Expertise and Information Technology*. University of Tampere, Department of Computer Science.
Verbeek, P.-P. (2006). Materializing Morality Design Ethics and Technological Mediation. *Science, Technology & Human Values*, *31*(3), 361–380.
Verbeek, P.-P. (2011). *Moralizing Technology: Understanding and Designing the Morality of Things*. University of Chicago Press.
Visser, F. S., Stappers, P. J., van der Lugt, R., & Sanders, E. B.-N. (2005). Contextmapping: experiences from practice. *CoDesign*, *1*(2), 119–149.
Wagner, I. (1993). A Web of Fuzzy Problems: Confronting the Ethical Issues. *Commun. ACM*, *36*(6), 94–101.
Winschiers-Theophilus, H., Bidwell, N. J., & Blake, E. (2012). Altering participation through interactions and reflections in design. *CoDesign*, *8*(2-3), 163–182.
Wölfel, C., & Merritt, T. (2013). Method Card Design Dimensions: A Survey of Card-Based Design Tools. In P. Kotzé, G. Marsden, G. Lindgaard, J. Wesson, & M. Winckler (Eds.), *Human-Computer Interaction – INTERACT 2013* (pp. 479–486). Springer Berlin Heidelberg.
Wood, J. (2007). *Designing for Micro-utopias; thinking beyond the possible*. Ashgate. Retrieved from http://research.gold.ac.uk/326/